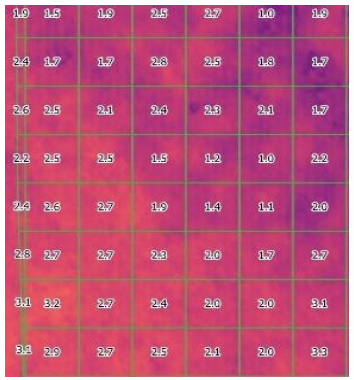
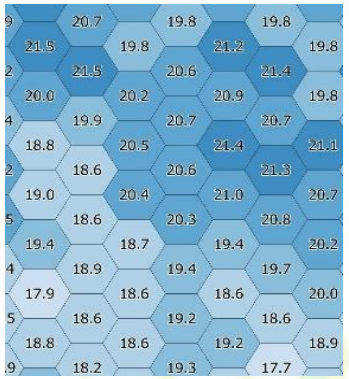
การตั้งเงื่อนไขการบินโดรนเพื่อให้ภาพถ่ายมีคุณภาพ

รศ.ดร.ขวัญตรี แสงประชาธนารักษ์
สาขาวิศวกรรมเกษตร คณะวิศวกรรมศาสตร์
มหาวิทยาลัยขอนแก่น







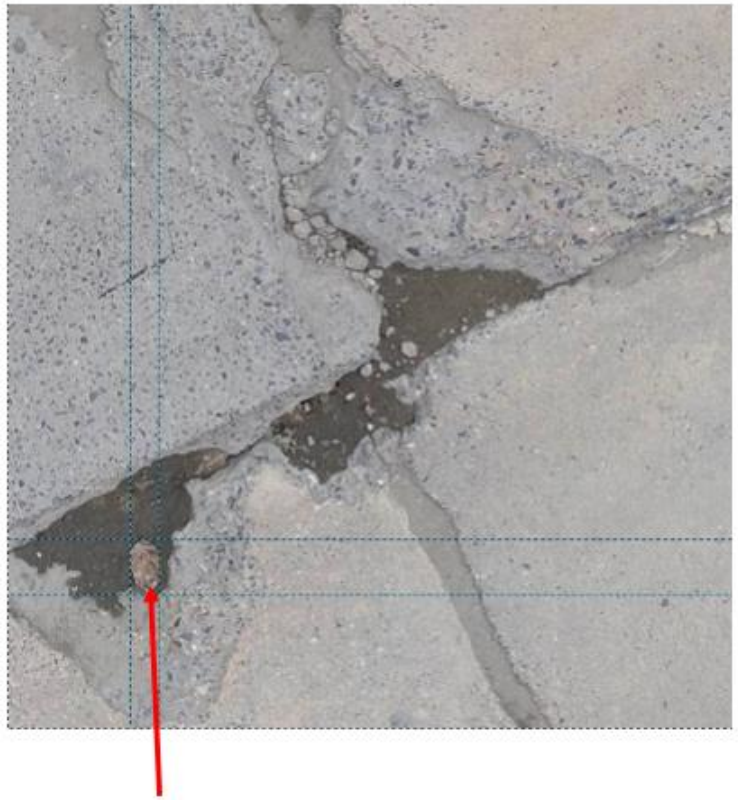




คุณภาพของภาพถ่าย

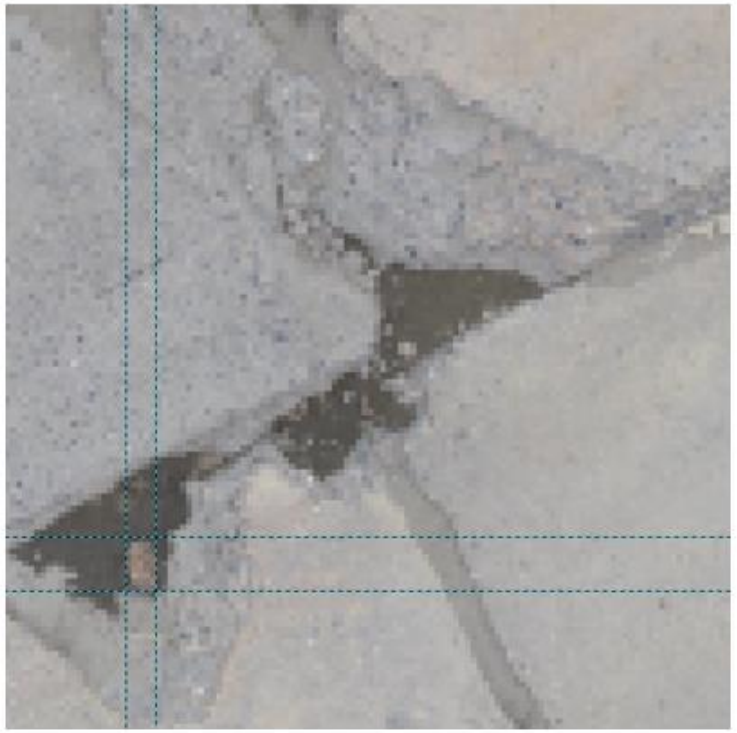
- Ground Sampling distance (GSD)
- Ground Control Point (GCP)
- Flight timing
- Drone traveling speed/Overlapping/Image scale
- Reflectance map/Calibration panel
- Low reflectance image
- Blur image
- Map Zone

Ground Sampling distance (GSD)



A leaf

- 46 x 73 px
- About 55.2 x 87.6 mm
- 1.2 mm GSD



10 mm GSD



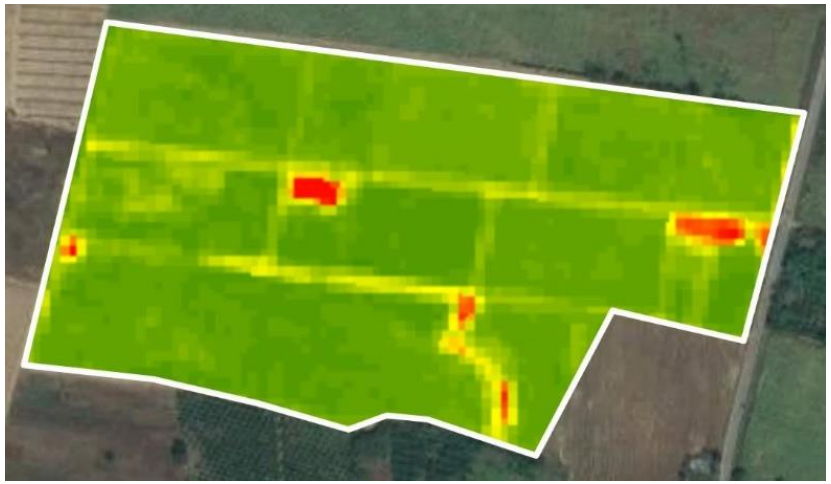
20 mm GSD

เปรียบเทียบระหว่าง ภาพจากดาวเทียมและโดรน

GPS: 13.6475662,102.3007576

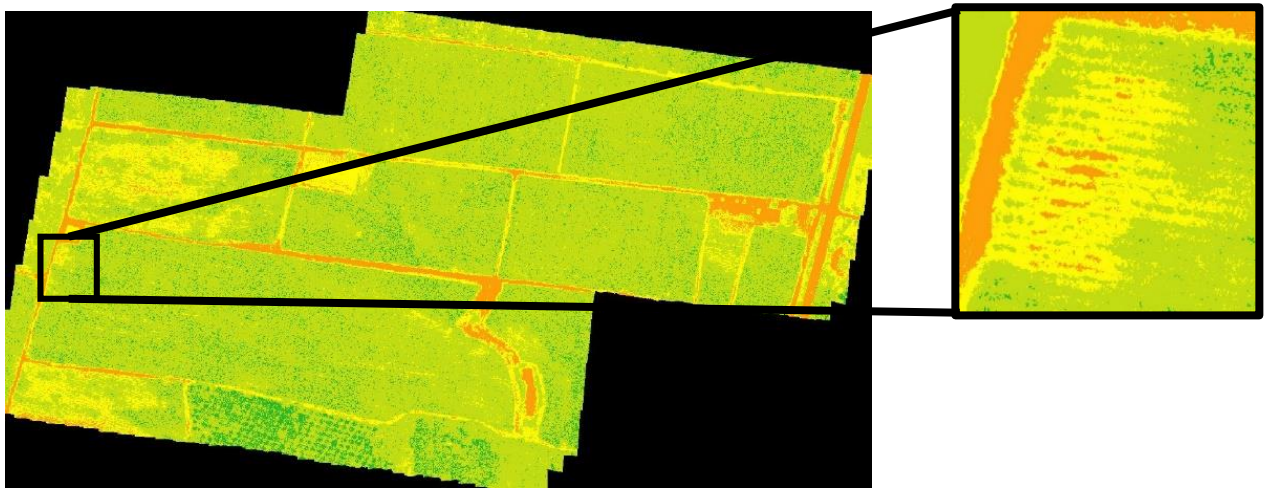
GSD 10 m/pixel



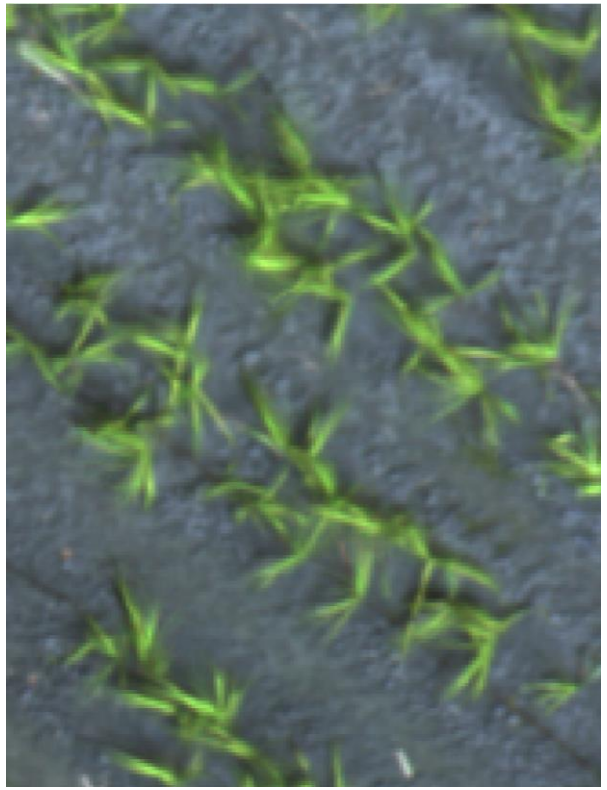


GSD 8 cm/pixel

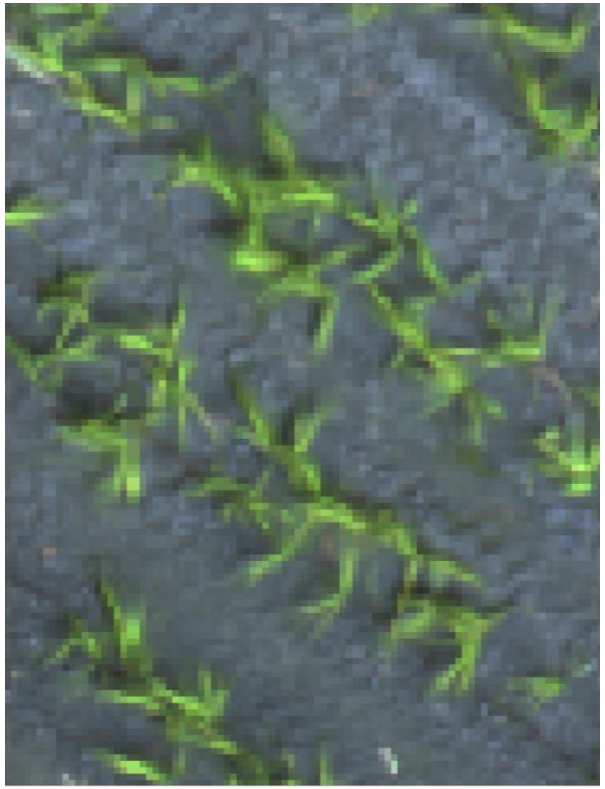




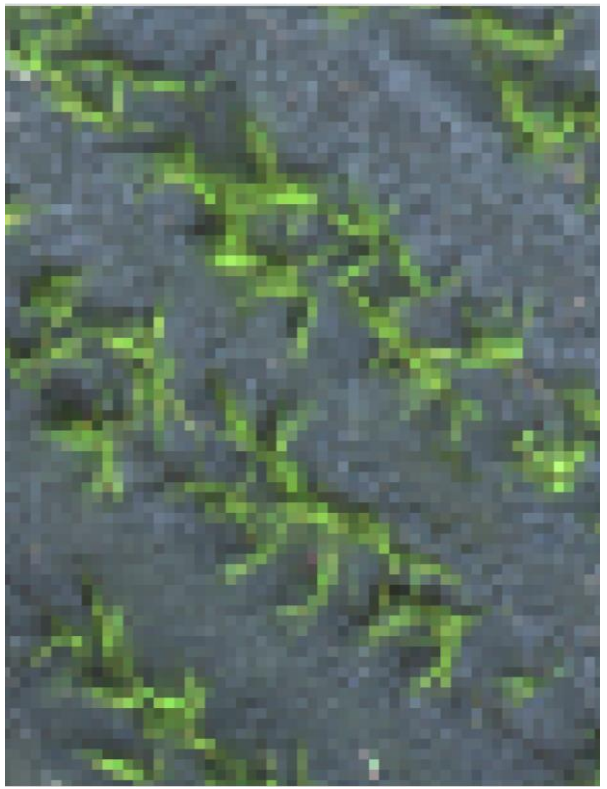
ภาพตัวอย่างความละเอียดต่างๆ



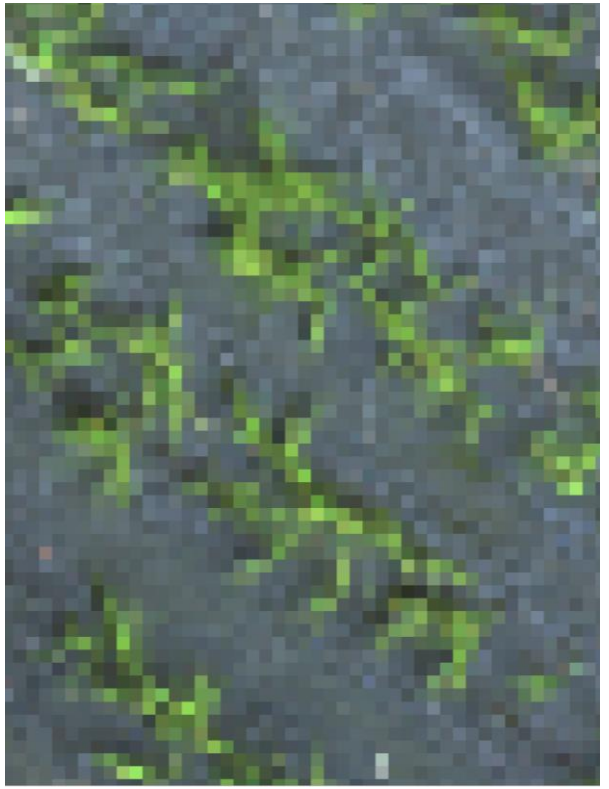
GSD 3 cm



GSD 5 cm



GSD 8 cm



GSD 10 cm

ไฟล์ใหญ่ขนาดไหน ?

- For a are around 10 ha or 62.5 Rai at 8 cm GSD
- RAW image is about 8 GB

100 MB / Rai

625 MB / ha

- 300,000 Rai -> 30TB/flight 

X 30

ส่งข้อมูลอย่างไร ?

- Approx 100 - 200 GB / day / drone
- 300 Mbps internet will take 2 hours
- 30 Mbps internet will take 20 hours
- 3 Mbps internet will take 200 hours

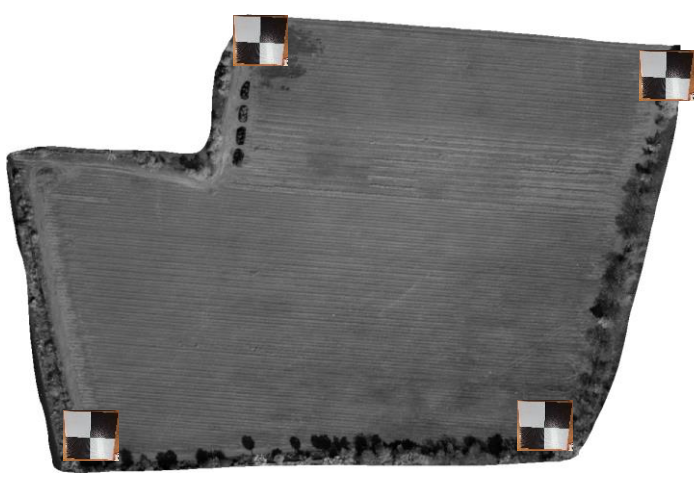


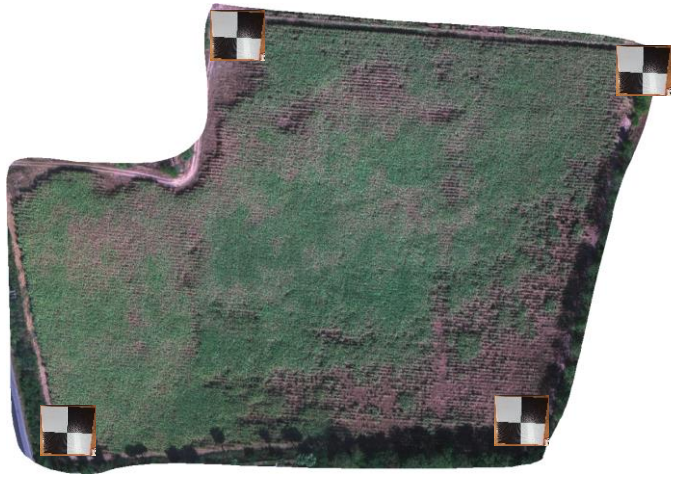


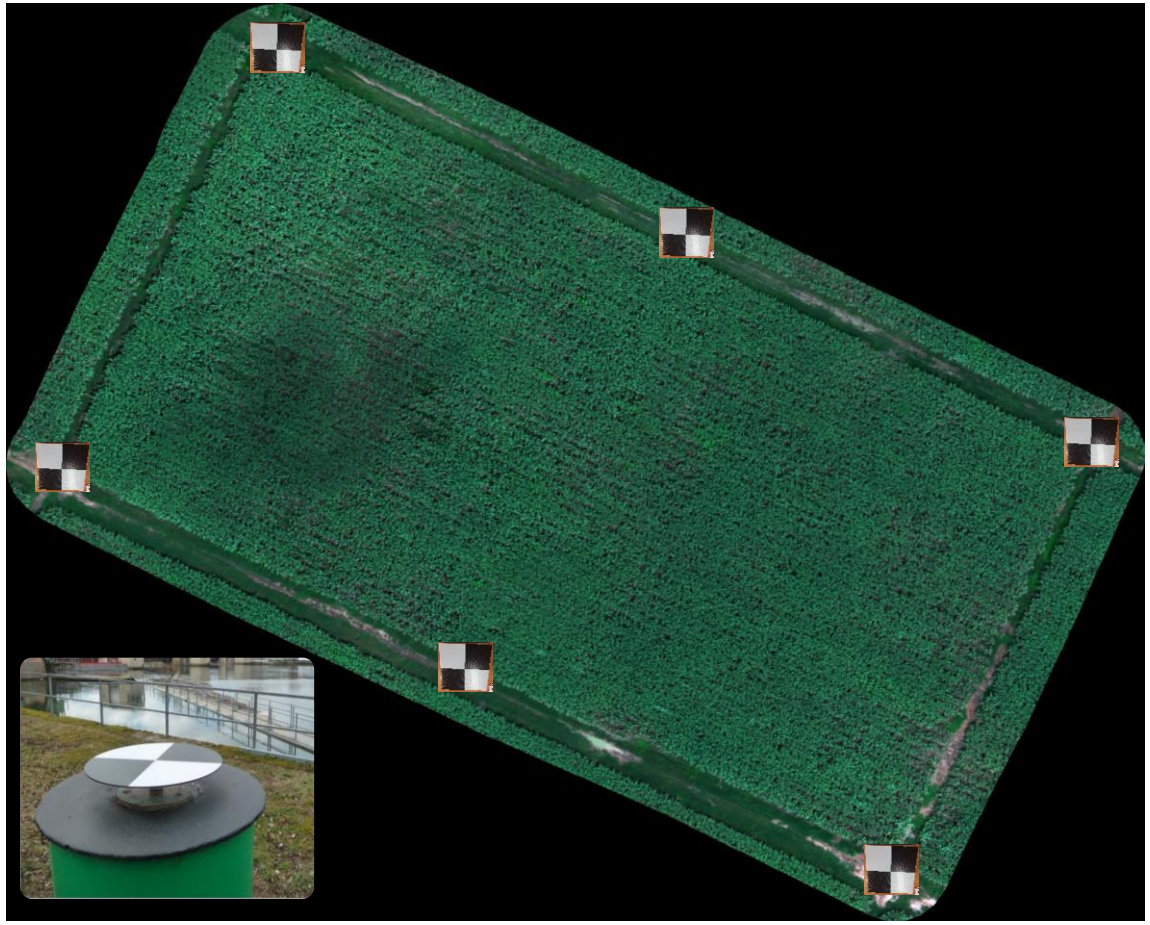
...?

Ground Control Point (GCP)

- ✓ ช่วยในการต่อภาพ โดยเฉพาะในแปลงอ้อยโตที่ยอดอ้อยปกคลุมพื้นที่ทั้งหมด
- ✓ ช่วยคำนวณความสูง โดยใช้ DSM – DTM (ใช้ร่วมกับ RTK GPS)





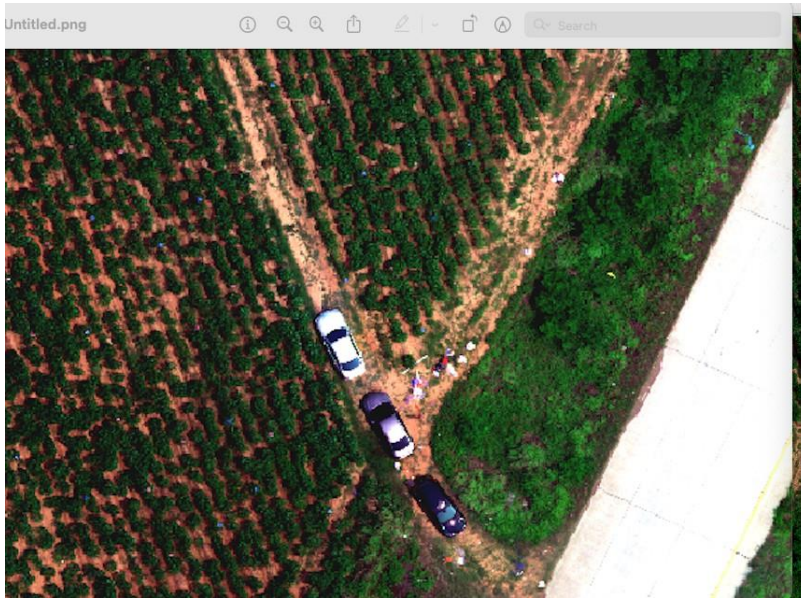


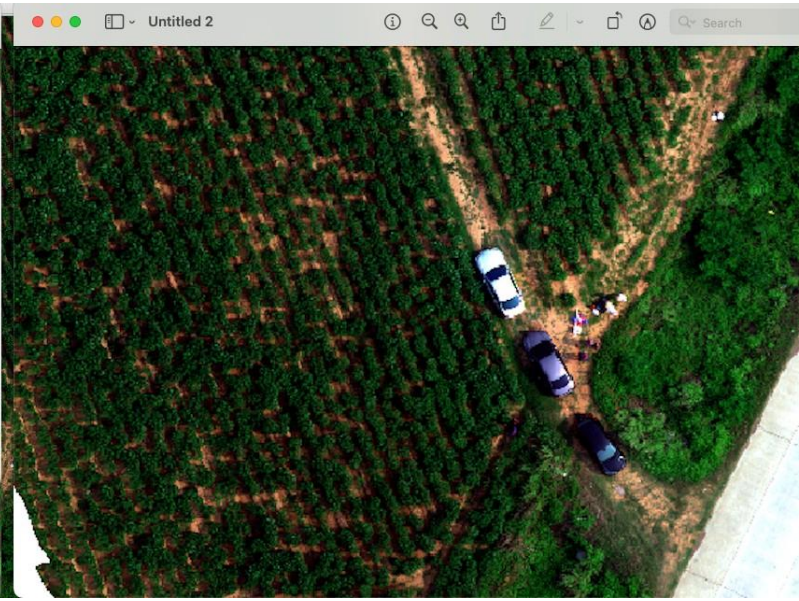
Flight timing

Shadow



Shadow





ปัญหาการต่อภาพไม่ได้/ ภาพแหว่ง (RGB camera)

➢Detail information of flight mission with error:

✓ Sony RX0, RGB and rolling shutter camera type

✓1 GSD equal to 24 meters above ground

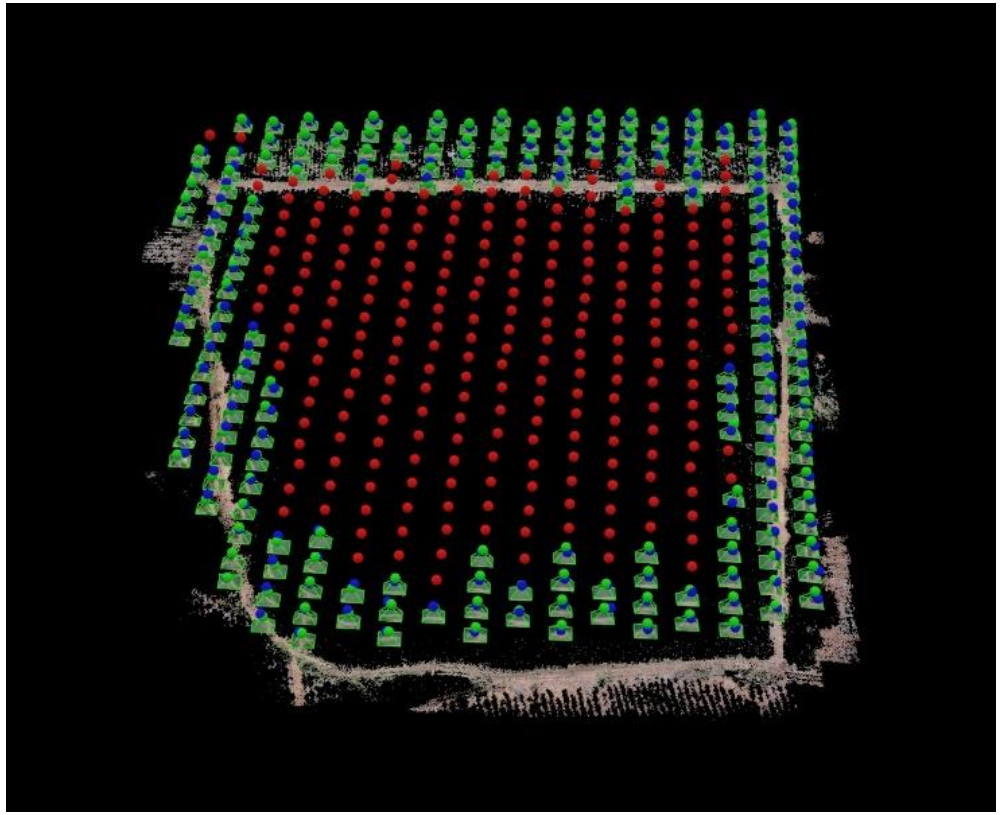
✓ Flight speed 5 meters per second

✓ Overlapping 75%

✓ Closed canopy sugarcane fields

✓ No GCPs or manual tie points

✓ Process in Pix4D with setting full image scale



ปัญหาการต่อภาพไม่ได้/ ภาพแหว่ง (RGB camera)

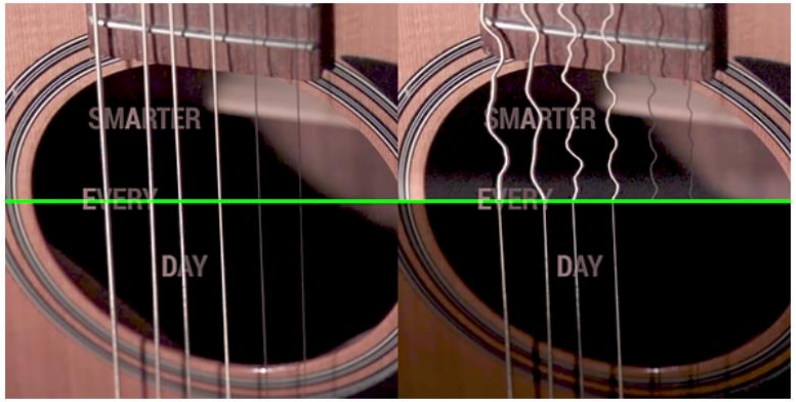
Camera suffers from very bad rolling-shutter effect

Fast-moving objects

Drove flight at high speed

Fly at low altitude

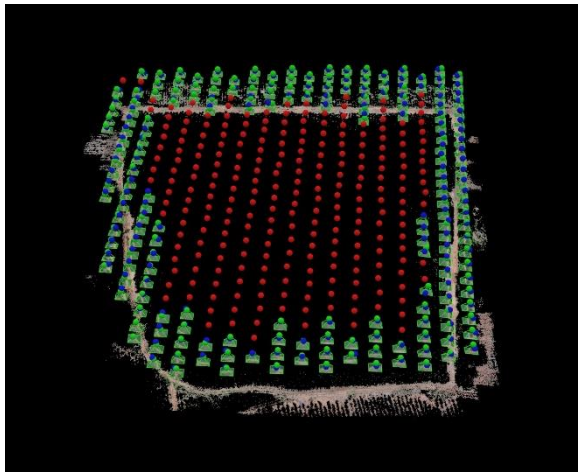
Images are blurry and leads to difficult to extract keypoints



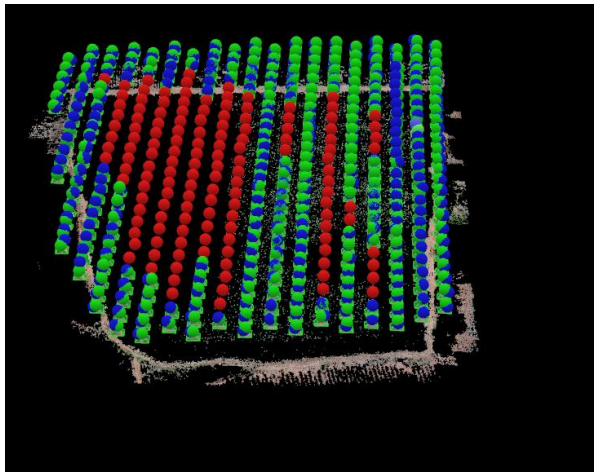
Rolling-shutter effect

ปัญหาการต่อภาพไม่ได้/ ภาพแหว่ง (RGB camera)

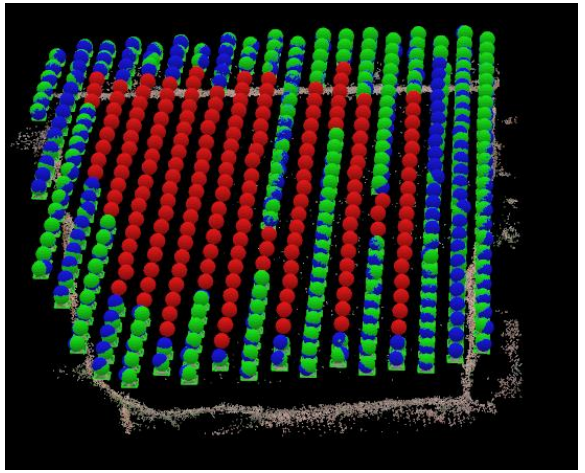
RGB camera stitching problems: Adjust process setting



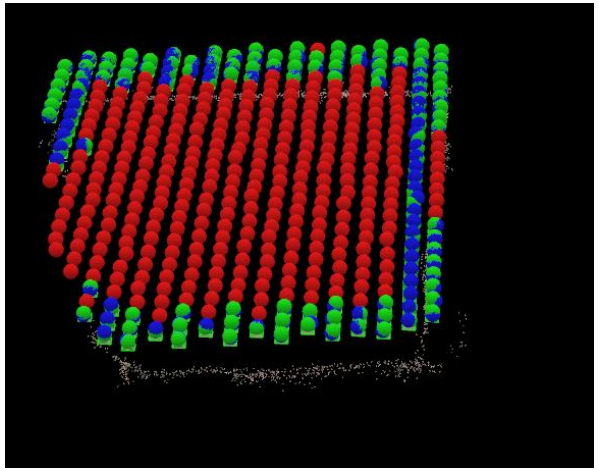
Full image scale



1/2 image scale



1/4 image scale

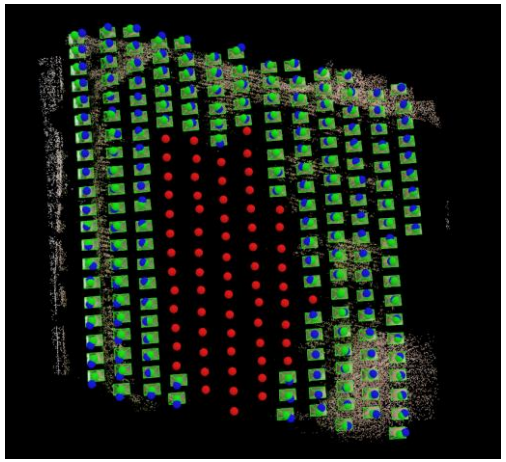


1/8 image scale

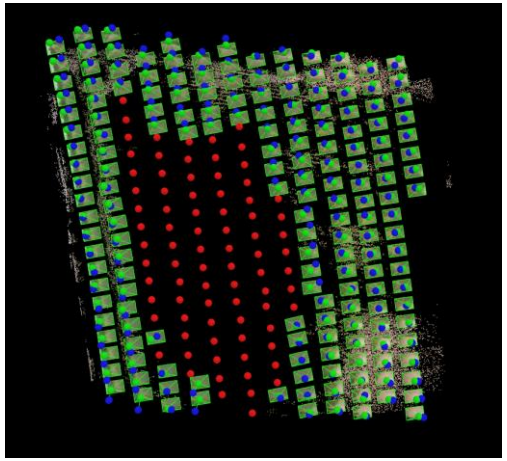
The best process setting for 1 GSD flight mission is ½ and ¼ image scales

ปัญหาการต่อภาพไม่ได้/ ภาพแหว่ง (RGB camera)

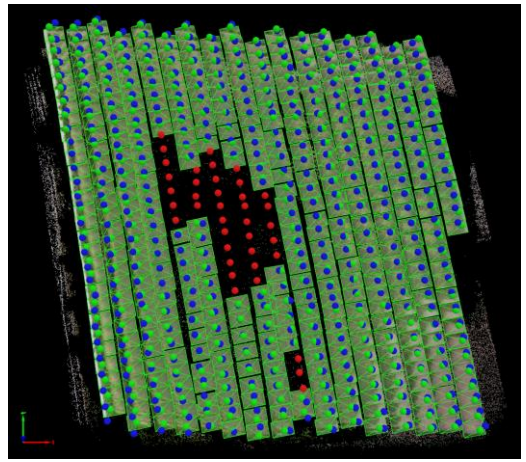
- ½ image scale



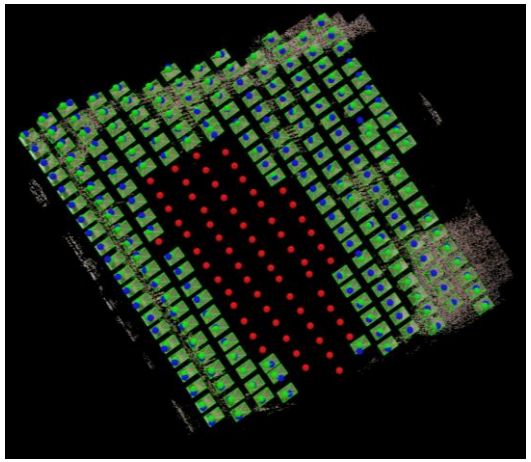
RGB camera, overlap 80%, speed 5ms



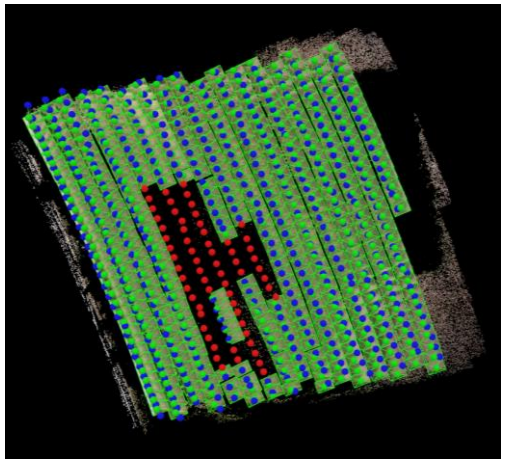
RGB camera, overlap 80%, speed 4ms



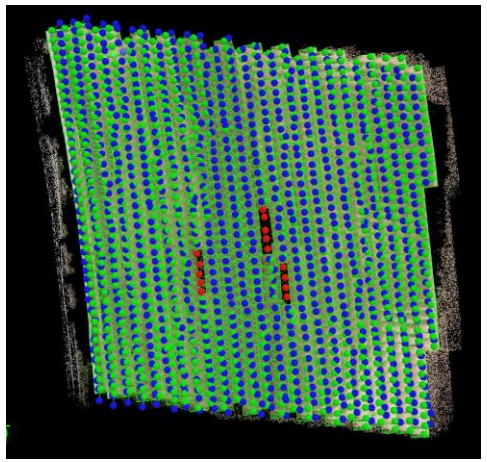
RGB camera, overlap 85%, speed 4ms



RGB camera, overlap 80%, speed 3ms



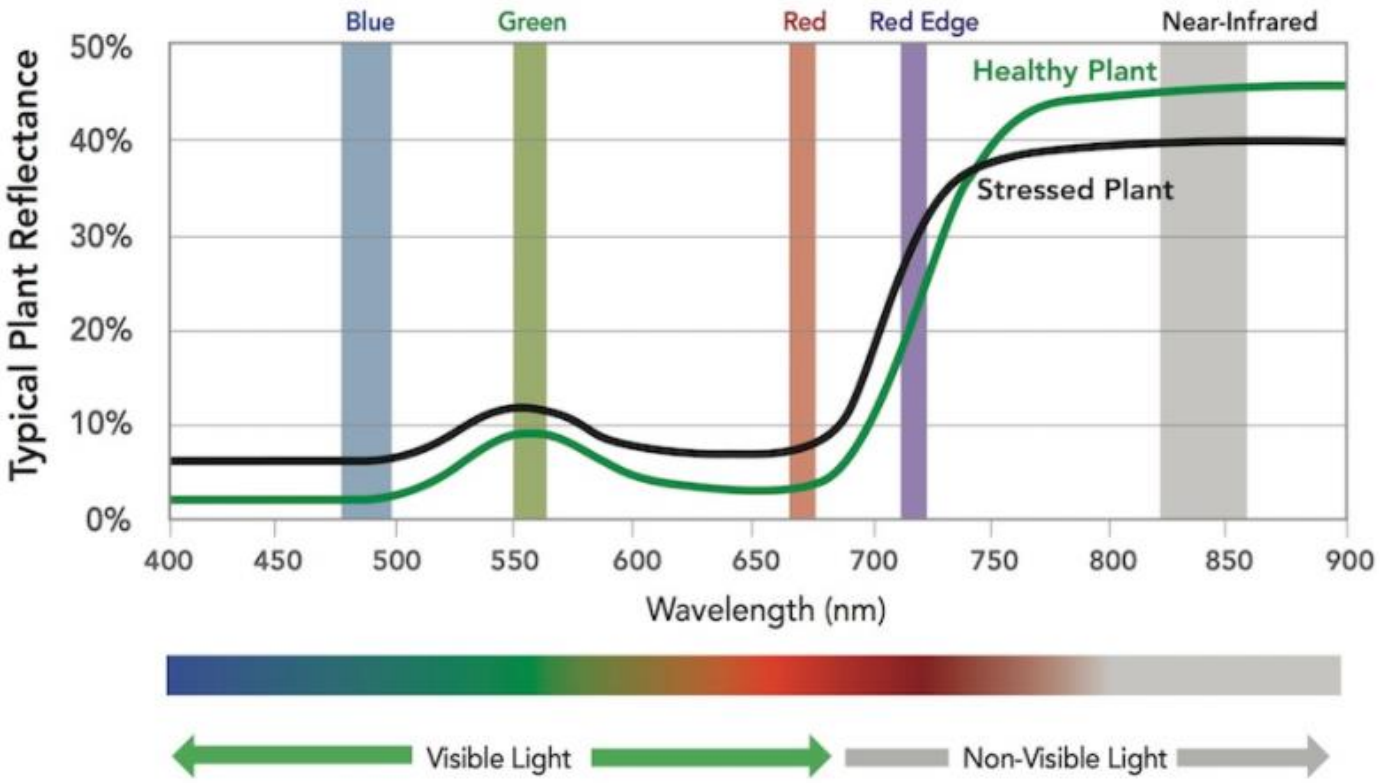
RGB camera, overlap 85%, speed 3ms

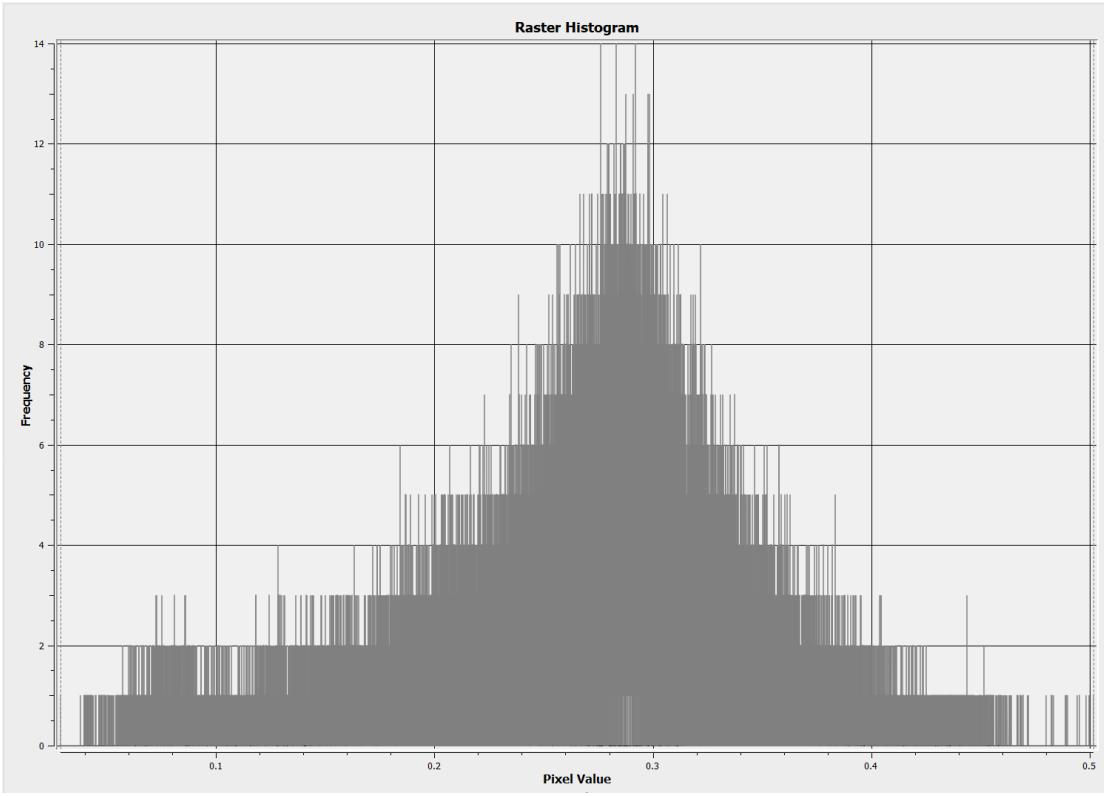


RGB camera, overlap 90%, speed 3ms

Reflectance

Reflectance is an inherent property of an object; it does not depend on lighting conditions. Its value range 0-1.





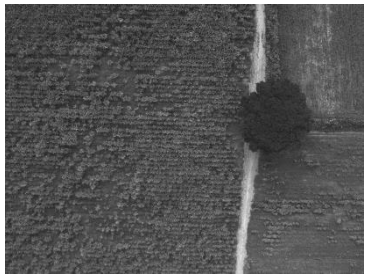
Reflectance value in NIR band

Reflectance

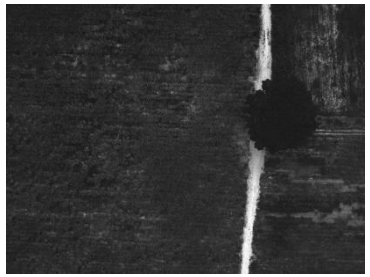
Raw pixel of Image is raw digital number from raw imagery, value range 0-255



Blue



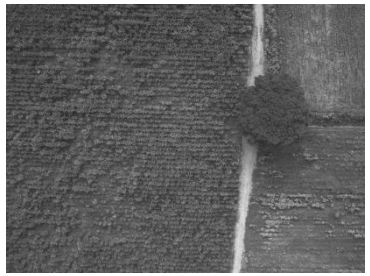
Green



Red



NIR



Rededge

Irradiance is solar irradiance, the light energy emanating from the sun as measured from the Earth after passing through the atmosphere




```
$ exiftool IMG_0106_2.tif
 ...
Irradiance : 0.96050399541854858
Irradiance Roll : 1.8932661497089052
Irradiance Pitch : -7.0249684496199967
Irradiance Yaw : 57.545878211823961
 ...
```

Downwelling Light Sensor (DLS)

- The spectral irradiance (W/m^2/nm) in each band sensed by the DLS for that band.
- The Irradiance Yaw, Pitch, and Roll provide the measured orientation of the DLS in degrees, are used to correct sun angle in Pix4D (option “Camera, sun irradiance and sun angle by DLS IMU”).

Reflectance

How to convert raw pixel of images to reflectance value?

Raw pixel of Images in Blue, Green, Red, NIR, Rededge band

+

Reflectance calibration factor

→

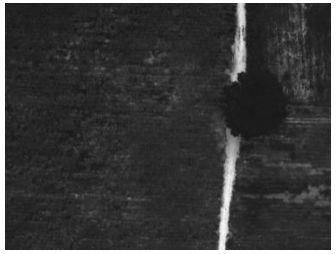
Reflectance value by Pix4D software



Blue



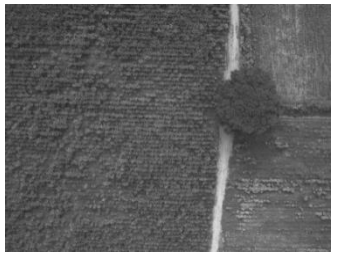
Green



Red



NIR



Rededge





DLS and sunshine sensor

$$F_i = \frac{\rho_i}{avg(L_i)}$$

where:

- F_i is the reflectance calibration factor for band i
- ρ_i is the average reflectance of the CRP for the *ith* band (from the calibration data of the panel provided by MicaSense)
- $avg(L_i)$ is the average value of the radiance for the pixels inside the panel for band i

Reflectance

How reflectance calibration factor during flight mission is extracted?

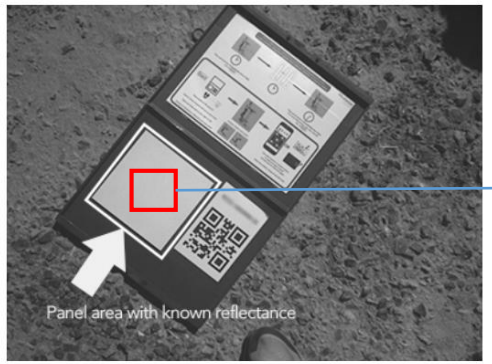
Raw pixels of calibrated reflectance panel

+

Know Reflectance curve

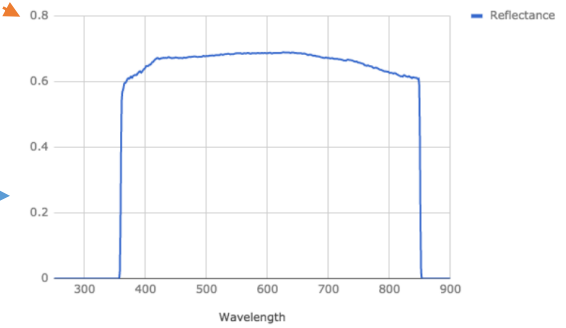
→

Reflectance calibration factor



Raw pixels

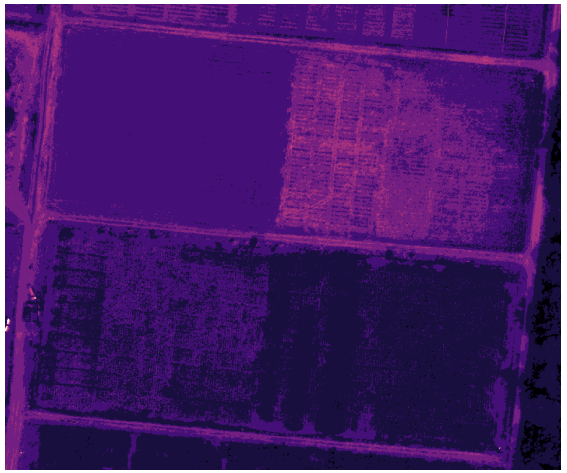
measured its reflectance across the spectrum of light captured by MicaSense Sensor.

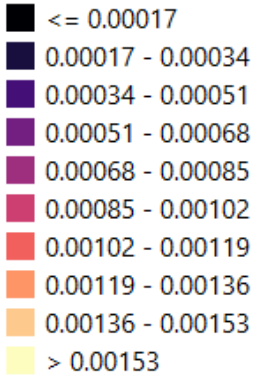


Known reflectance curve

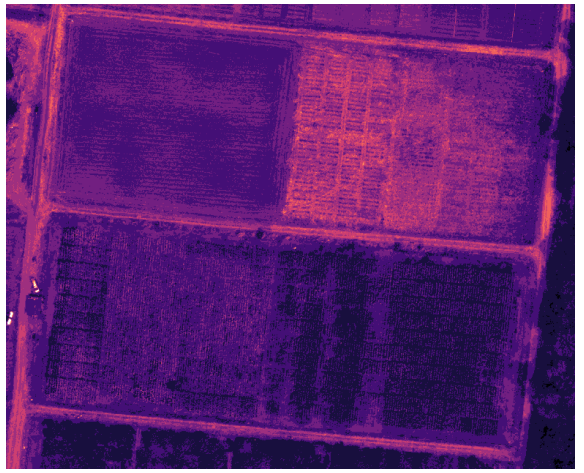
- So, that is why selecting the images of calibrated reflectance panel is very important because **if images of calibrated reflectance panel were taken under different light condition from during flight mission, it will cause low reflectance value**.

Reflectance

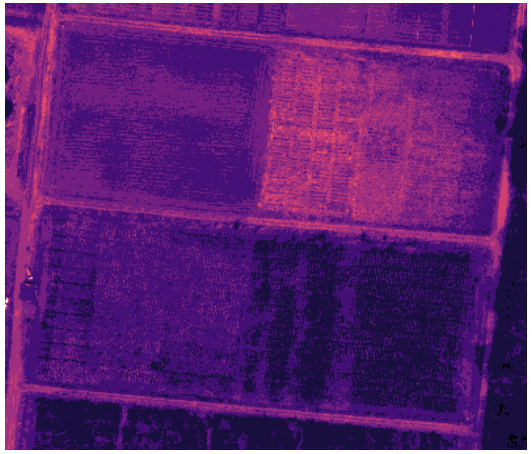


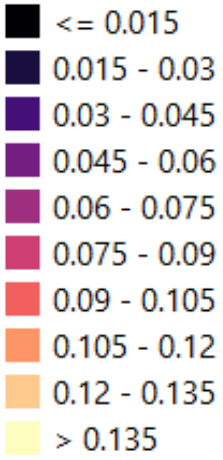


Blue band, Rededge camera, no panel

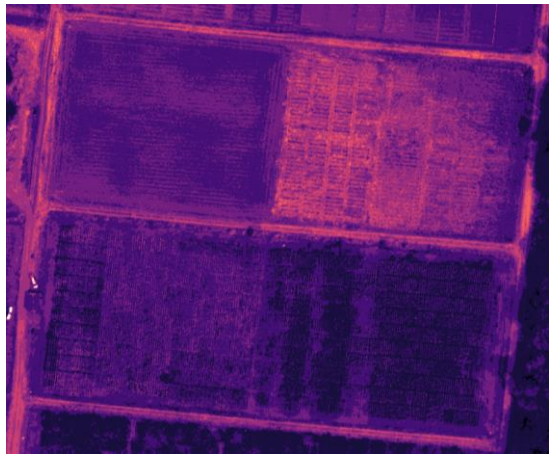


Blue band, Rededge MX camera, no panel



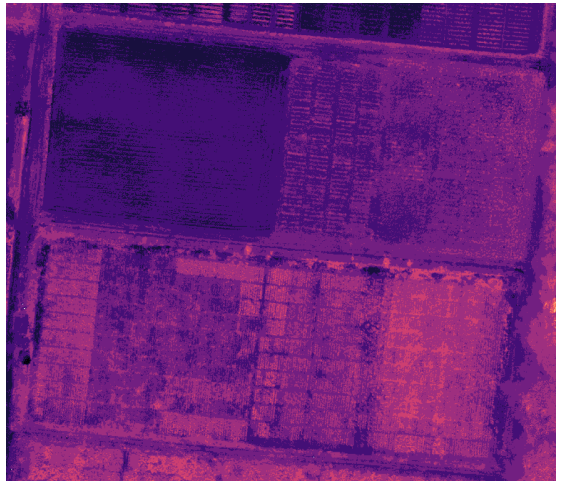


Blue band, Rededge camera, have panel

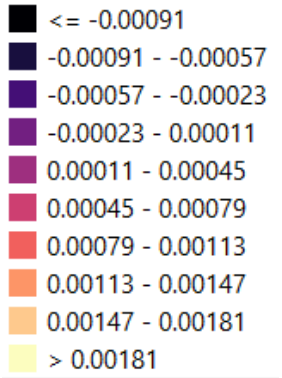


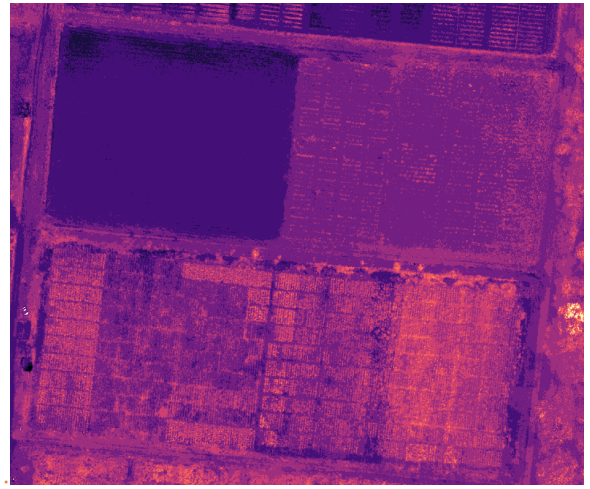
Blue band, Rededge MX camera, have panel

Reflectance

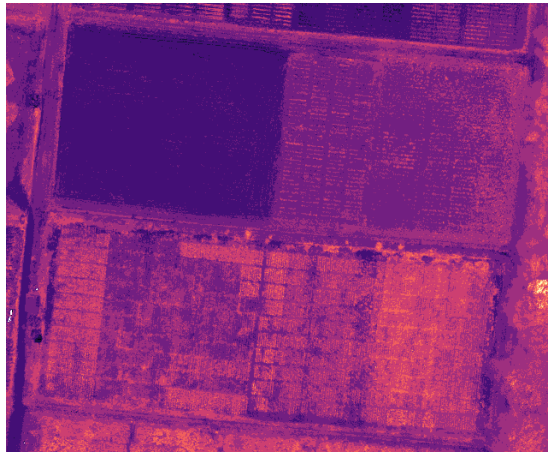


ExG, Rededge camera, no panel

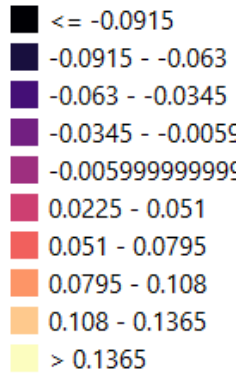


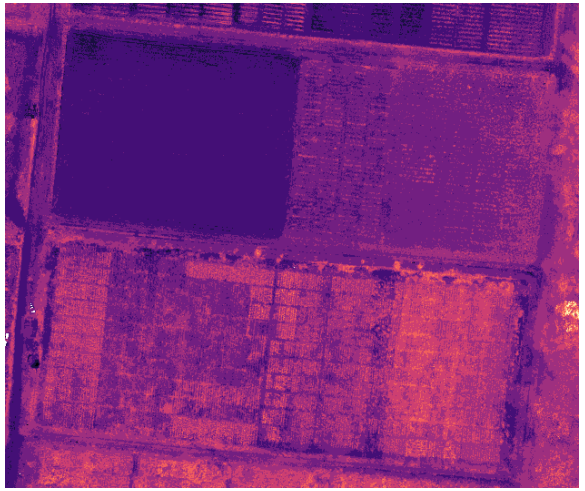


ExG band, Rededge MX camera, no panel



ExG band, Rededge camera, have panel





ExG band, Rededge MX camera, have panel

Difficulties to get stable light during flight mission

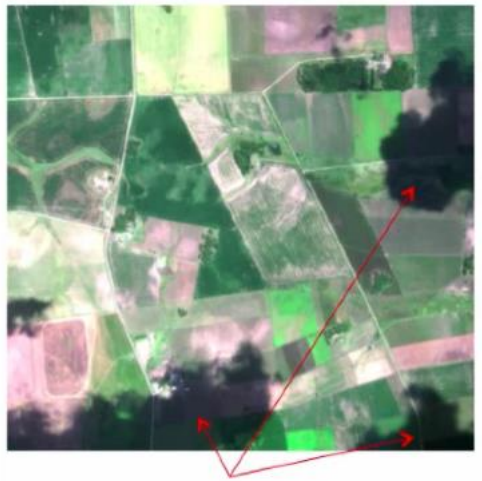




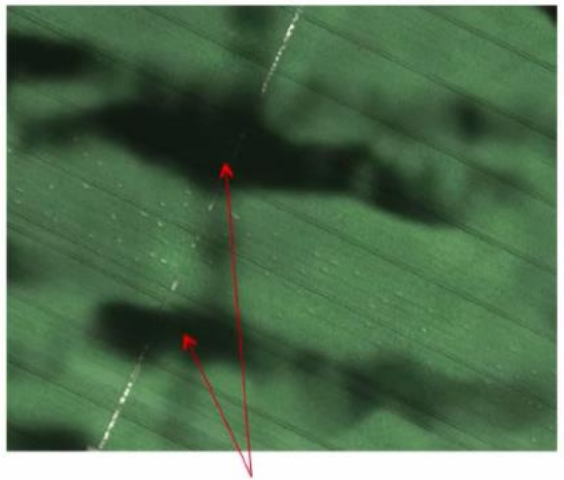
Drone can flight in average about 30 minutes

During 30 minutes flight mission

For 30 minutes flight mission, ambient light can change anytime due to cloud and wind



Clouds



Clouds Passing By

Reflectance

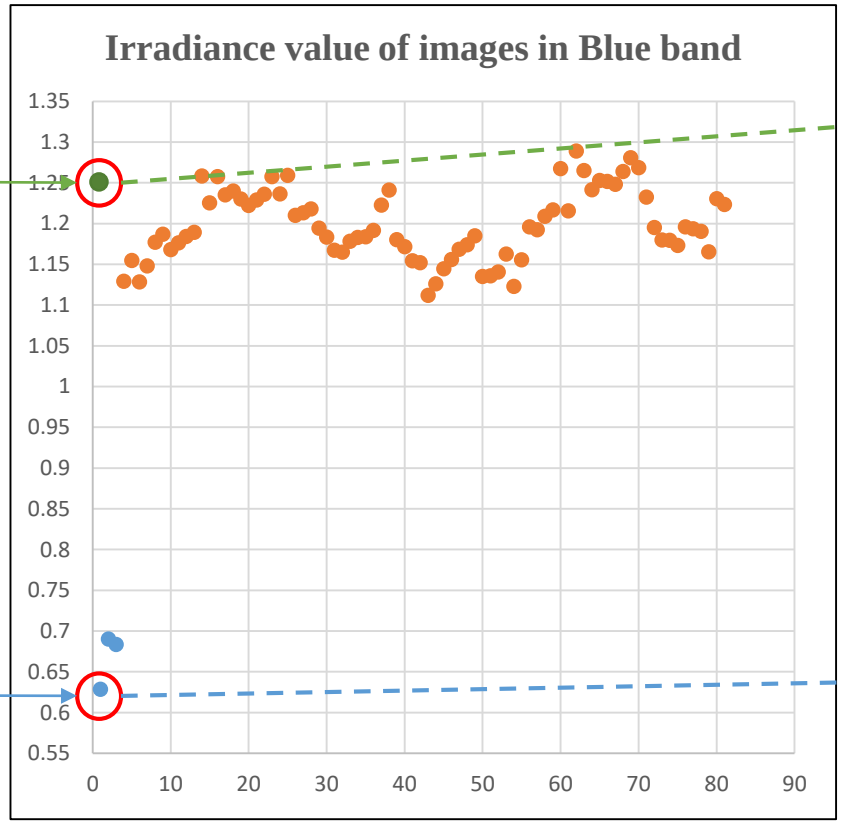
Image of Calibrated Reflectance Panel (CRP) and irradiance value and low and correct reflectance map

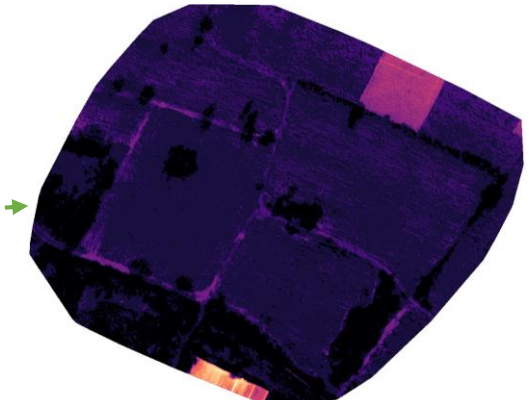


- Image of CRP in blue band, irradiance value 1.25
- Taken **from different day**

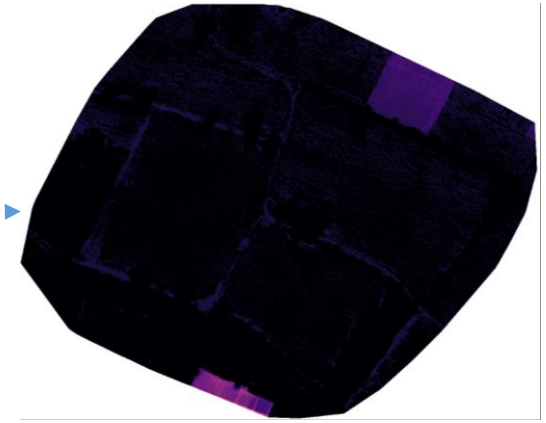


- Image of CRP in blue band, irradiance value 0.62
- Taken **before flight mission**

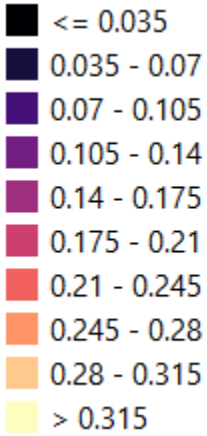




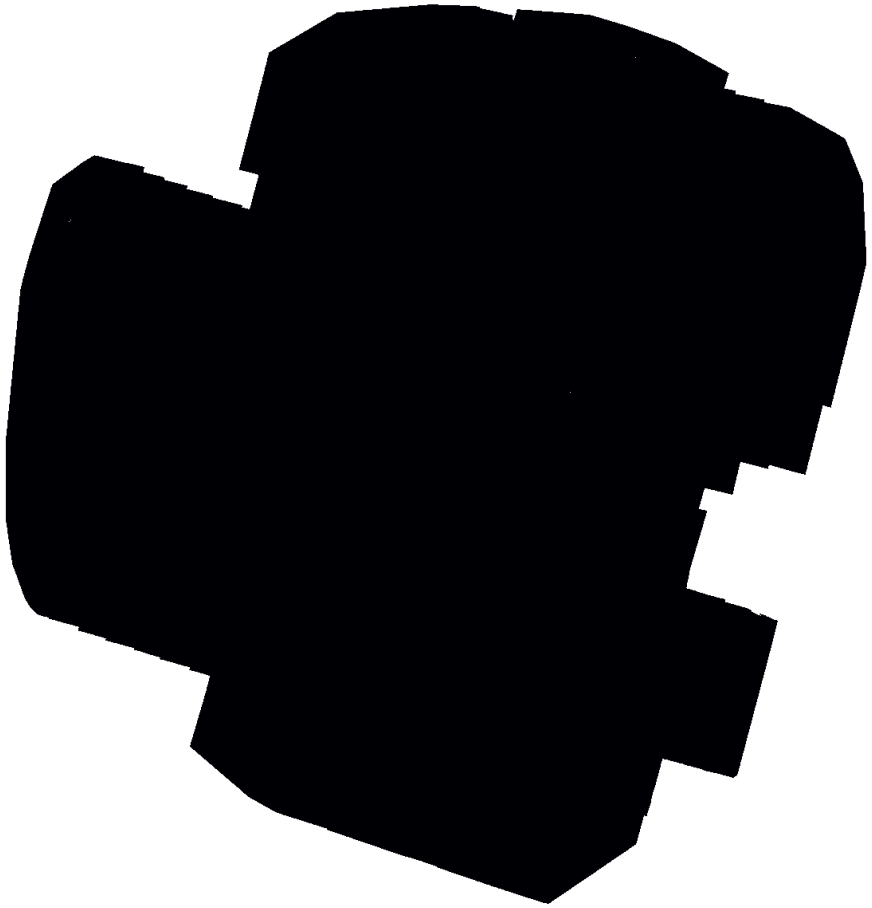
Correct reflectance map in Blue band

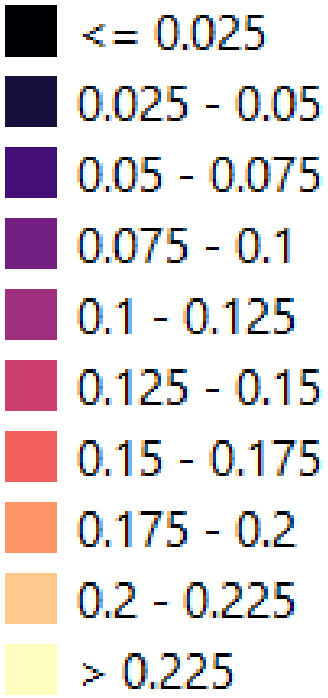


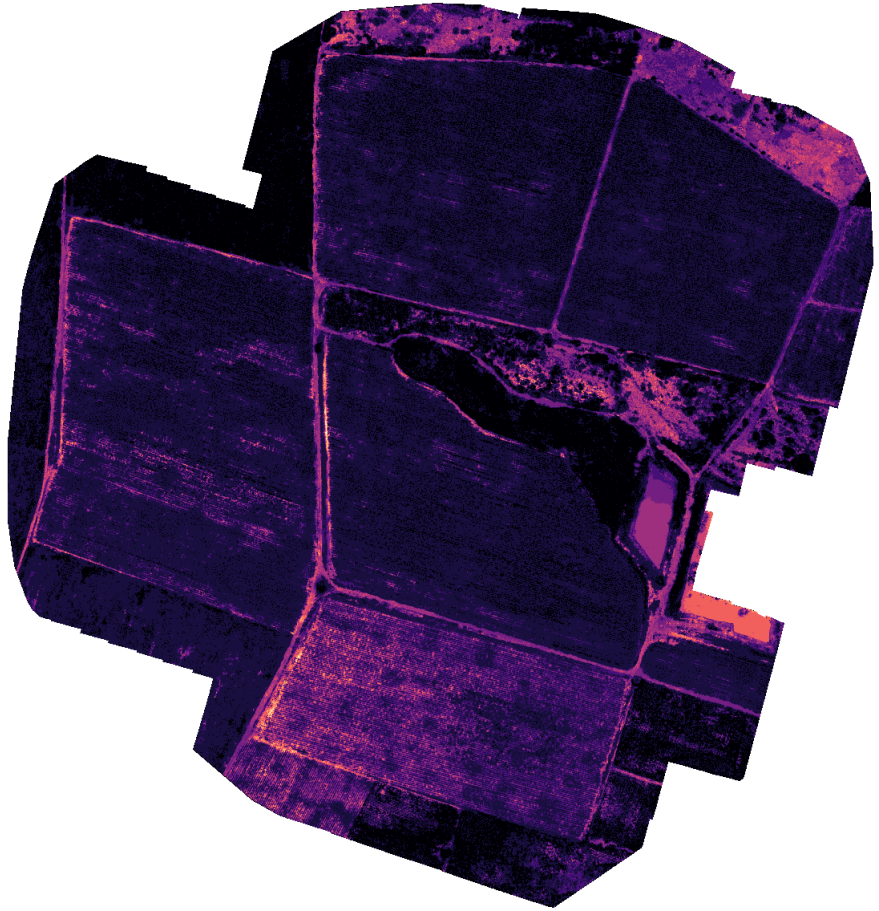
Low reflectance map in Blue band



Low and correct reflectance in Blue band





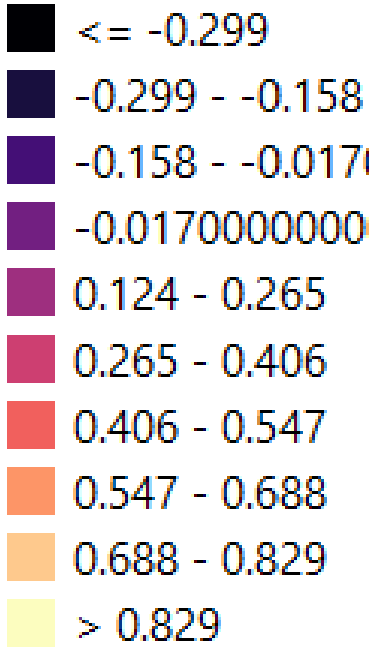


Low reflectance

Correct reflectance

NDVI map calculated from low and correct reflectance



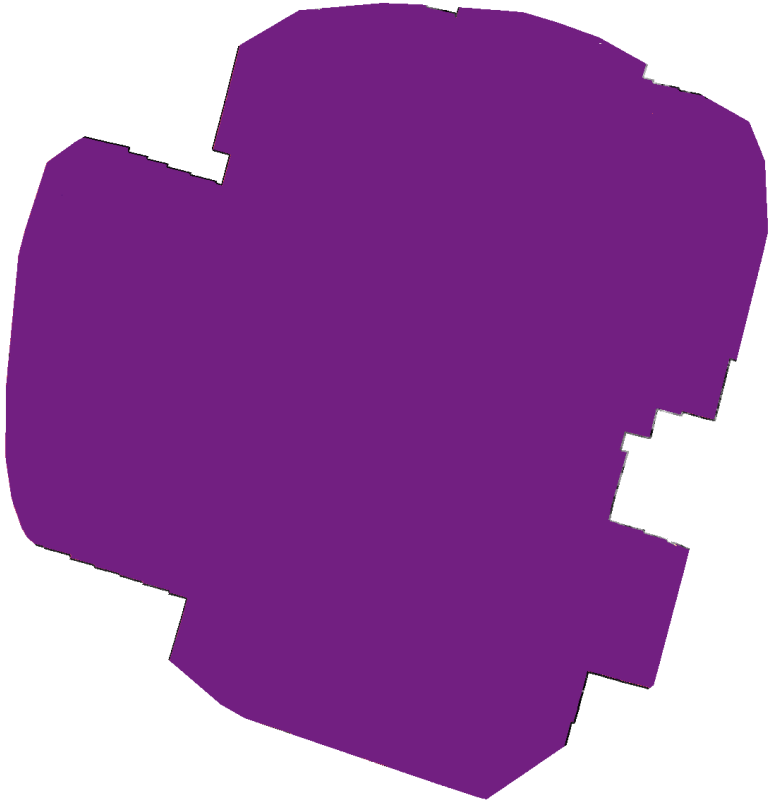


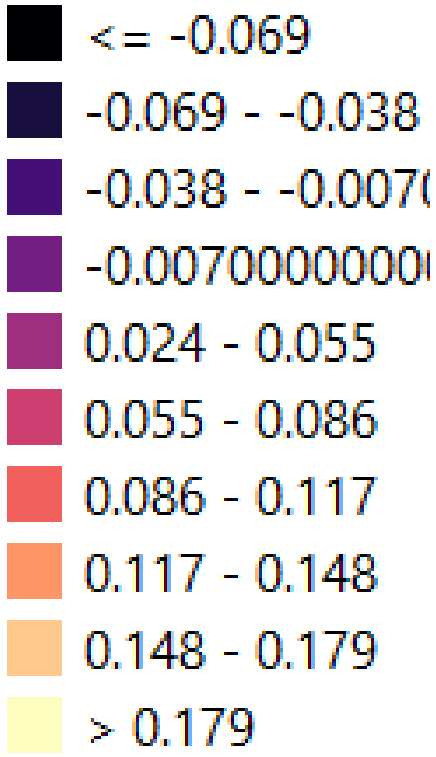


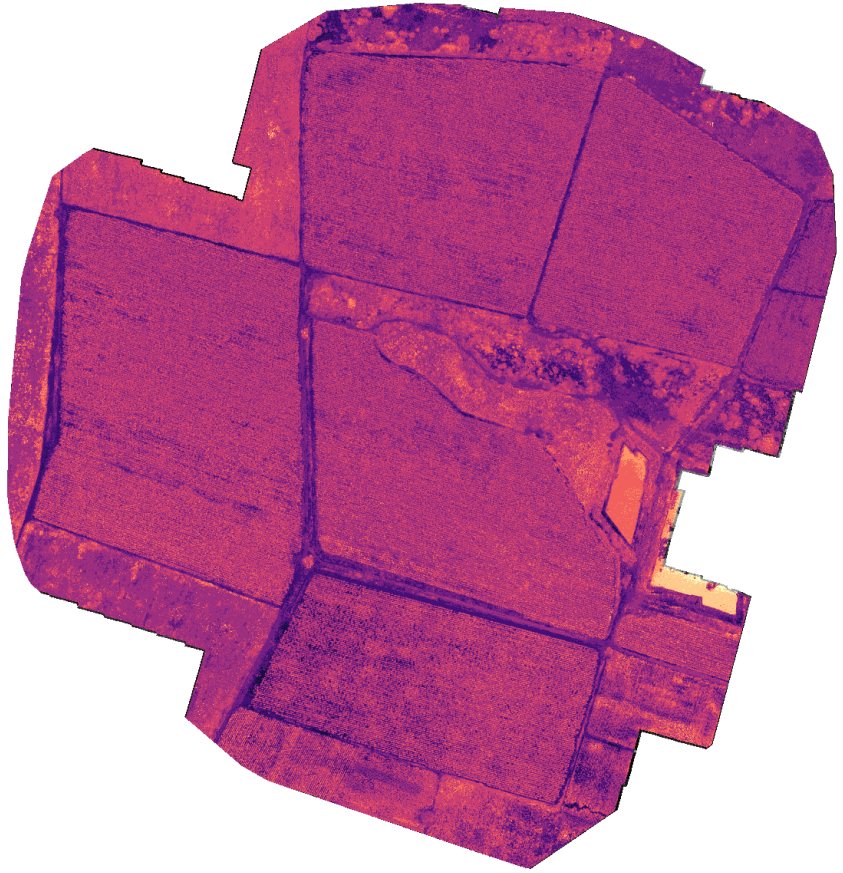
Low reflectance

Correct reflectance

ExG map calculated from low and correct reflectance







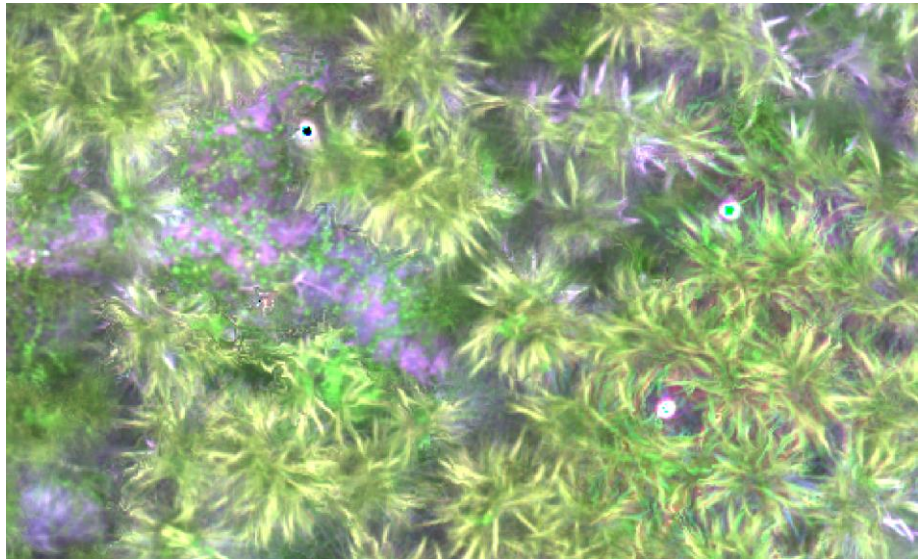
Low reflectance

Correct reflectance

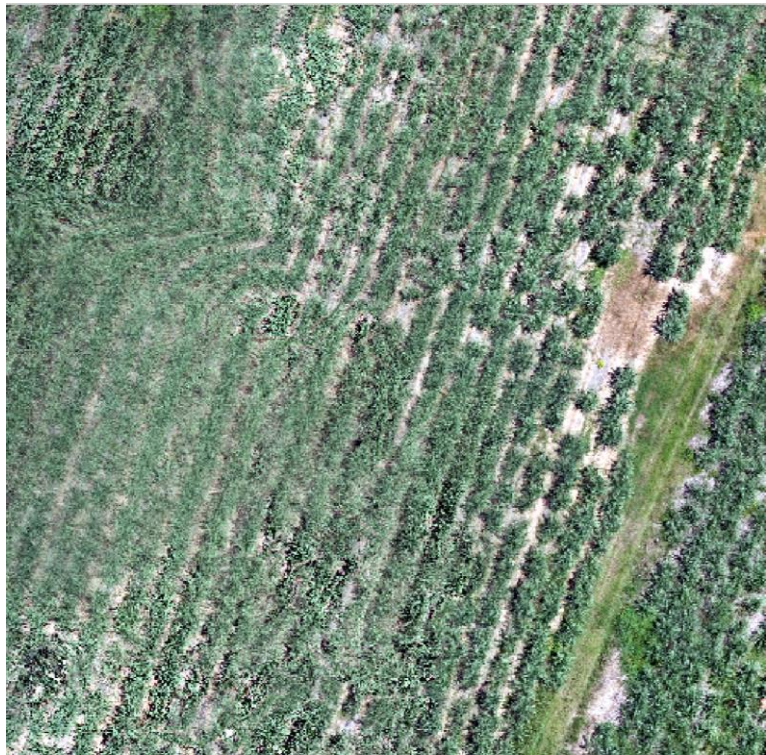
ตัวอย่างแผนการบินที่ต่อภาพไม่ได้ ภาพมัวเนื่องจากเป้าขยับ ใบอ้อยไหว ที่ความละเอียดสูง



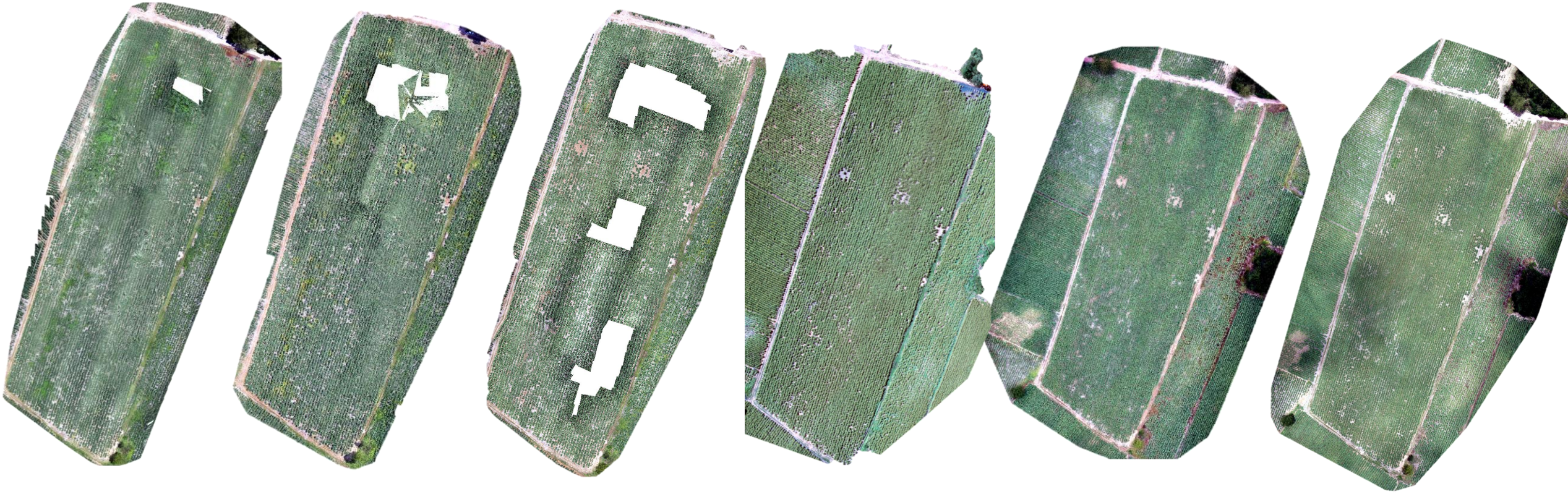








ตัวอย่างแผนการบินที่บินแผนเดิมแต่รูปร่างแปลงและสภาพแปลงมีความแตกต่างกัน



1 day

7 day

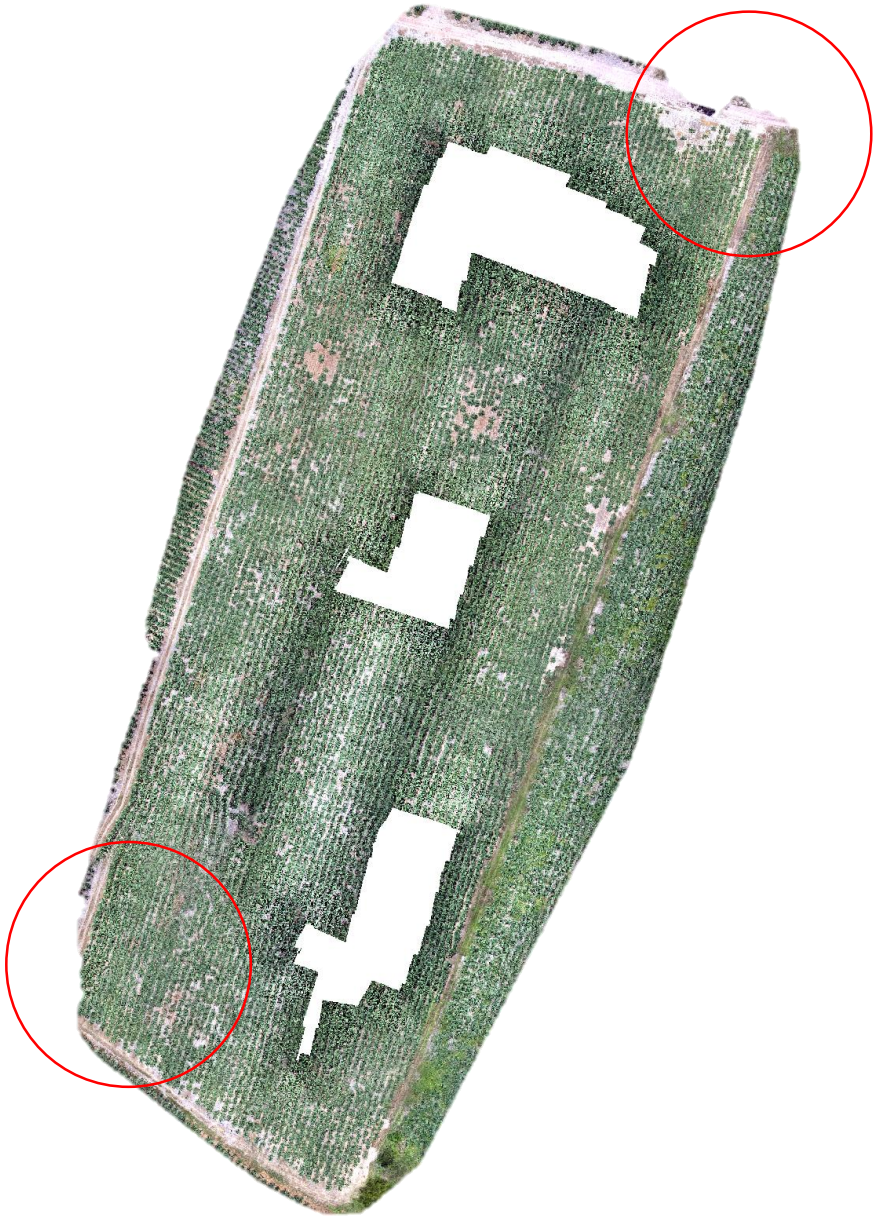
14 day

21 day

28 day

45 day

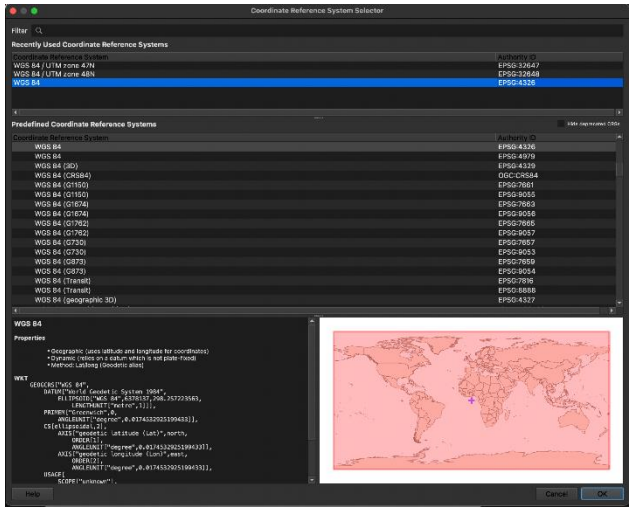
ตัวอย่างแผนการบินที่ตำแหน่งในการสร้างแผนการบินอาจจะเลื่อน

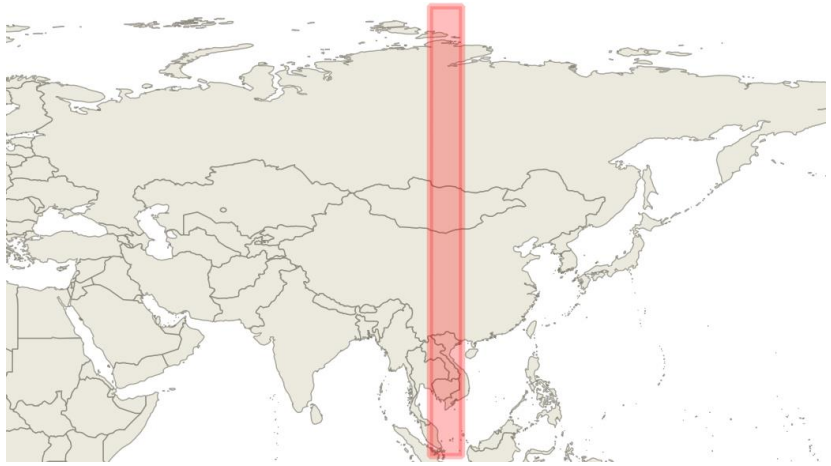


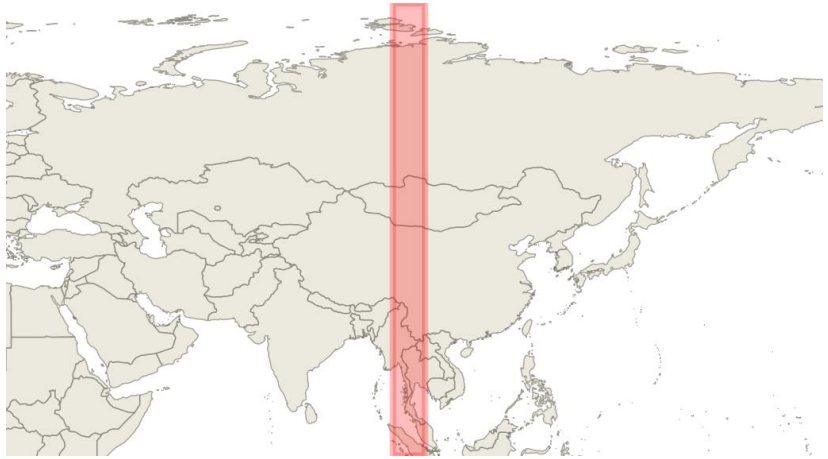




โซนแผนที่







Information from provider

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GDAL Driver Metadata	GeoTIFF
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Information from provider

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GDAL Driver Description	GTiff
GDAL Driver Metadata	GeoTIFF
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Q&A
Post-test